

Embedded Linux & System Software Interview Handbook

Phase 2 – Linux Memory & Process Management

This phase combines two of the most important Linux kernel topics for senior Embedded Linux, BSP and Kernel Developer interviews.

Chapter 8 – Linux Memory Management

Topics Covered:

- Virtual vs Physical Memory
- Address Space Layout
- MMU (Memory Management Unit)
- Paging and Page Tables
- Translation Lookaside Buffer (TLB)
- Kernel Memory Layout
- Buddy Allocator
- SLAB / SLUB / SLOB Allocators
- kmalloc(), kcalloc(), vmalloc(), vzalloc()
- DMA Memory Allocation
- High Memory & Low Memory
- Page Cache
- Copy-on-Write (CoW)
- Anonymous Memory
- Memory Mapping (mmap)
- Huge Pages & Transparent Huge Pages
- NUMA
- Memory Compaction
- Memory Fragmentation
- OOM Killer
- kmemleak, slabtop, /proc/meminfo
- Real Driver Examples
- Interview Questions

Chapter 9 – Linux Process Management

Topics Covered:

- Process vs Thread

- Task Structure (task_struct)
- Fork, vfork, clone
- exec() Family
- Context Switching
- Linux Scheduler
- Completely Fair Scheduler (CFS)
- Real-Time Scheduling
- Scheduling Policies
- CPU Affinity
- Signals
- Process States
- Kernel Threads
- Wait Queues
- IPC Mechanisms
- Pipes, FIFOs
- Shared Memory
- Message Queues
- Semaphores
- Namespaces
- cgroups
- Process Debugging (ps, top, strace, perf)
- Interview Questions

Learning Outcomes

After completing this phase you will understand how Linux manages memory and schedules processes, how kernel allocators work, how virtual memory is translated by the MMU, how the scheduler selects tasks, and how to debug memory and process-related issues in production systems.

Phase	Status
Phase 1	Completed (Chapters 1–7)
Phase 2	Memory Management & Process Management
Estimated Size	~250–350 pages when fully expanded
Target	Senior Embedded Linux / Kernel Interviews